

Customer Churn Prediction & Analysis

1. Introduction

Customer churn refers to the loss of customers who stop using a company's product or service. Predicting churn is very important for businesses because retaining existing customers is more cost-effective than acquiring new ones.

This project focuses on analyzing customer behavior and predicting churn using Machine Learning and visualizing insights using Power BI.

2. Objectives

- To understand customer behavior and patterns
 - To identify factors influencing customer churn
 - To build a Machine Learning model for churn prediction
 - To create an interactive dashboard for business insights
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3. Dataset Description

The dataset used in this project is the **Telco Customer Churn Dataset**.

◆ Features include:

- Customer demographics (Gender, Senior Citizen)
 - Account information (Tenure, Contract Type)
 - Services used (Internet, Phone, etc.)
 - Billing details (Monthly Charges, Total Charges)
 - Target variable: **Churn (Yes/No)**
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4. Data Preprocessing

The following steps were performed:

- Converted TotalCharges to numeric format
- Handled missing values using median imputation
- Dropped unnecessary columns like customerID
- Converted target variable (Churn) into binary (0/1)
- Applied one-hot encoding to categorical variables

5. Exploratory Data Analysis (EDA)

EDA was performed to understand patterns and relationships:

Key Visualizations:

- Churn Distribution
- Monthly Charges vs Churn
- Tenure vs Churn
- Contract Type vs Churn
- Internet Service vs Churn

Observations:

- Customers with **month-to-month contracts** have higher churn
- **Higher monthly charges** are associated with higher churn
- Customers with **low tenure** are more likely to churn

6. Model Building

Algorithm Used:

- Logistic Regression

Steps:

- Split data into training and testing sets
- Applied feature scaling using StandardScaler
- Trained the Logistic Regression model

7. Model Evaluation

- Accuracy achieved: **~82%**
- Confusion Matrix used to evaluate performance

Interpretation:

- Model performs well in predicting churn and non-churn customers
- Provides a good baseline for business decision-making

8. Dashboard (Power BI)

An interactive dashboard was created to visualize insights:

◆ **Features:**

- KPI Cards (Total Customers, Churn, Churn Rate)
- Interactive filters (Gender, Contract, Internet Service)
- Multiple charts for trend analysis

◆ **Insights:**

- Month-to-month customers churn the most
 - Fiber users show higher churn tendency
 - Churn rate is approximately **26%**
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💡 **9. Business Insights**

- Retention strategies should focus on new customers
 - Pricing strategies may impact churn
 - Long-term contracts help reduce churn
 - Personalized offers can reduce customer loss
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🚀 **10. Conclusion**

This project successfully demonstrates how Machine Learning and data visualization can be used together to analyze and predict customer churn.

The insights generated can help businesses make data-driven decisions to improve customer retention and reduce churn.

🔧 **Tools & Technologies**

- Python (Pandas, NumPy, Scikit-learn)
 - Power BI
 - Jupyter Notebook
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📌 **Future Scope**

- Use advanced models like Random Forest or XGBoost
- Deploy the model using a web application

- Perform hyperparameter tuning for better accuracy
- Integrate real-time data for continuous prediction

 Author

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